

Introduction to Tokens and Tokenization

Spoken Tutorial

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<https://spoken-tutorial.org>

Learning Objectives

In this tutorial, we will learn,

- ▶ **Explain what tokens are, how tokenization works, and why tokens are crucial for LLM performance, cost, and efficiency.**



System Requirements

- ▶ **Here I am using a browser on a computer or mobile.**
- ▶ **We will also use an online tokenizer tool.**

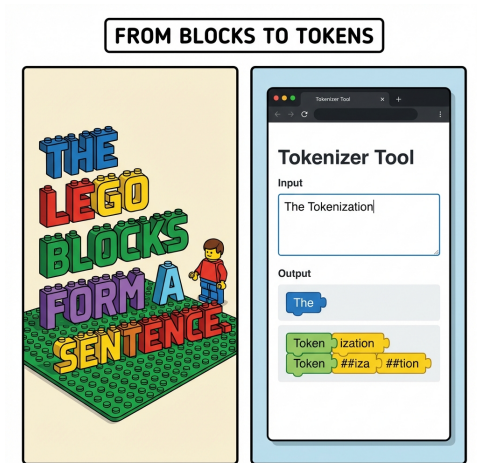


Pre-requisite Slide

- ▶ **Introduction to Generative AI,
Introduction to Prompting**



What are Tokens?

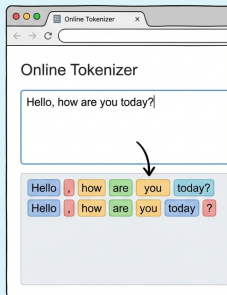


How Tokenization Works

Analogy:
Chopping into pieces



Reality:
Tokenization process



Why Tokens Matter: Context Windows

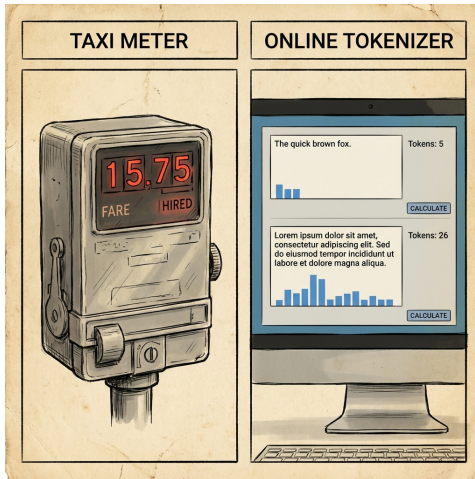
- ▶ Context window is LLM's short-term memory
- ▶ Measured in a fixed number of tokens

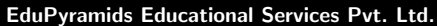


Text exceeding



Why Tokens Matter: Cost





Summary

In this tutorial, we learned that:

- ▶ **With this, we come to the end of this tutorial.**
- ▶ **Tokenization is the process of splitting text into these...**
- ▶ **And tokens are critical for context windows, cost, and...**



Assignment

- ▶ Find an online tokenizer tool.
- ▶ Experiment with different types of text: short words, long...
- ▶ Observe how the token count changes.
- ▶ Try to find a sentence that results in exactly 10 tokens.



Domain Expert
(Domain Expert)
Affiliation

Thank you

